

$\hat{\alpha}$ $\hat{\beta}$ $\hat{M} \rightarrow$ irrelevant δ $\alpha: t \rightarrow \Delta X$
 $\hat{\beta}$ $\hat{M} \rightarrow$ $x \rightarrow \Delta T$
 β

strat of 1 till 2m strat of 3
 strat of 2

2015

Microeconomics II
Final exam

Please solve the exam individually.

1. In standard signaling games, every Weakly Perfect Bayesian Equilibrium (WPBE) is a Sequential Equilibrium (SE).

Consider the following variation of a standard signaling game. Nature chooses $t \in T$ using the probability distribution ρ . Player 1 observes t and chooses a message $x \in X$. Player 2 observes the message x chosen by player 1 and responds with $y \in Y$. Player 3 observes the message x chosen by player 1 and responds with $z \in Z$. (Note that player 2 does not observe player 3's action when choosing y , and player 3 does not observe player 2's action when choosing z .) The distribution ρ and payoff functions are assumed to be common knowledge.

Is every WPBE also a SE in the game described?

Evaluate. (While a proof or counterexample, as appropriate, is the first-best response, a clear statement properly justified suffices.)

2. Consider a symmetric independent private values model with I bidders. For each bidder $i \in I$, i 's valuation x_i is distributed as a uniform in $[0, 1]$.

The seller uses the following auction. Bidders indicate their bids independently and simultaneously in a sealed envelope. The seller collects the envelopes. The highest bidder wins the object. In case of a tie, the object is assigned with equal probability among the bidders who tie. All bidders pay their own bids regardless of whether they get the object.

- a. Prove that the strategy $b(x_i) = Kx_i^S$ is a Bayesian Nash equilibrium for some values of the constants K and S . Identify the corresponding values of K and S .

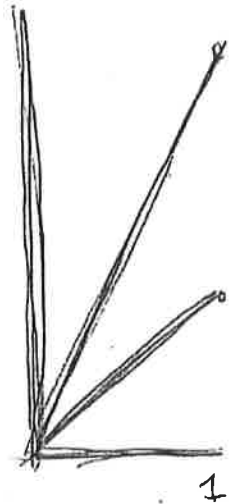
- b. Evaluate: The auction described is better for the seller than the second price auction in terms of expected seller's revenue.

3. Consider the following instance of the "market for lemons." Each seller knows, privately, the quality $x \in [0, 1]$ of her car, x is also the seller's reservation price. Potential buyers assign value ax , $a > 1$ to a car of quality x . (Potential buyers do not observe the quality of the car.) If a transaction takes

gets x , pays Kx^S
2 players

$$\int_0^1$$

$$\frac{x^{I+1}}{I+1} \Big|_0^1 = \frac{1}{I+1}$$



$$\min \left\{ \frac{a}{2} (2-r^2), \frac{a}{2} (1+r^2) \right\}$$

to gains = $ax - p + p - x$

$$\int_0^1 ax - p dx = \frac{ax^2}{2} \Big|_0^1 - p \times 1$$

place at a price p then, for an item of quality x , the seller's payoff is $p - x$ and the buyer's payoff is $ax - p$. If a transaction does not take place, buyer and seller receive payoff zero. It is common knowledge that x is distributed according to the cumulative distribution function $F(x) = x$.

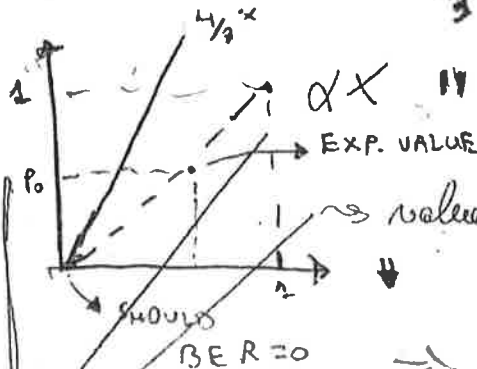
Suppose that sales of items with qualities less than $r \in [0, 1]$ is prohibited by law. Thus, no seller can offer an item whose quality is $x < r$.

- Given $a, p > 0$, and r , with $p > r$, what is the expected value to the buyer of a randomly picked car brought to the market.
- If $r = 0$, what values of a lead to the "Akerlof's result," i.e. no trade except with the lowest quality car.
- What happens if $a = 2$ and $r = 0$? $\Rightarrow \text{EXP VAL} = 1 \Rightarrow p = 1$
ALL CARS SOLD $\Rightarrow a = 2p$
- Given a and $p > 0$, determine the minimum value of r such that the market clears and some legal trades occur.
- Suppose $a = 4/3$. What is the value of r and the market clearing price p that maximize the aggregate gains from trade.

$$\begin{aligned} &= \frac{a}{2} - \frac{a}{2} r^2 \\ &+ p - pr \\ &= \frac{a}{2} (1 - r^2) - p(1 - r) \\ &= \frac{a}{2} (1 - r^2) - p(1 - r) \\ &= \frac{a}{2} (1 - r^2) - p(1 - r) \end{aligned}$$

$$x < 1$$

$$x^2 < x$$



$$\frac{2}{3} (1 - r^2) - p(1 - r)$$

$$p \geq 1$$

$$\frac{2}{3} (1 - r^2) = p(1 - r)$$

$$\frac{2}{3} (1 - r)(1 + r) = p(1 - r)$$

$$\Rightarrow 1 + r = \frac{3}{2} p$$

$$\Rightarrow r = \frac{3}{2} p - 1$$

$$p = \frac{2}{3}$$

$$a = 1.5 \quad p = 2$$

FROM ①: $\frac{a}{2} (1 - r^2) = p(1 - r)$

$$\Rightarrow p = \frac{a}{2} (1 + r)$$

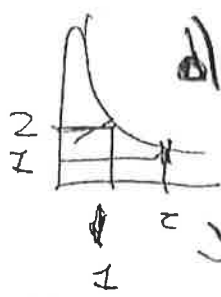
ONLY TRADE THIS TYPE: $p = r$

$$\frac{a}{2} (1 - r^2) = p \cdot \frac{a}{2} r = 1 + r$$

$$\frac{a}{2} (1 - r^2) = r \Rightarrow 2 - 2r = 1 + r$$

$$\frac{a}{2} = \frac{2}{3}$$

$$a = \frac{4}{3}$$



$$p = 2/2$$

$$a = \frac{4}{3}$$

